DIGITAL TWIN HACKATHON / HAPPIEST HEALTH

Gluco Twin

A metabolic digital twin that predicts your glucose before you eat.

TEAM HACKZZ / IIT (BHU), VARANASI

One diet does not fit all

100M+

Indians live with diabetes, and a similar number with prediabetes.

UNIQUE

Glucose response shifts with sleep, stress, BMI, age and meal makeup.

REACTIVE

CGMs are costly and show only what has already happened.

GAP: people need a safe way to test lifestyle choices before acting on them.

Why a digital twin, and why now

Twin thinking

A persistent virtual copy of you for safe experiments.

Cheap sensing

Phones and watches already track sleep, steps and stress.

Mature AI

Small ML models predict individual responses, even in a browser.

Prevention

Catching prediabetes early costs far less than treating diabetes.

A virtual you for what-if meals

- Enter your profile: age, BMI, fasting glucose, sleep, stress
- Describe a meal: carbs, protein, fat, fiber
- The twin predicts your 3-hour glucose curve
- It tests a better plan: fewer carbs, more fiber, a 15-minute walk
- You see both curves and the predicted peak drop

How it works

INPUTS

Profile, meal, activity

FEATURES

10 model features

AI MODEL

Gradient Boosting

TWIN ENGINE

Curve + what-if

DASHBOARD

Streamlit + web

FULL DIAGRAM: docs/architecture.pdf

Model details

Model

Gradient Boosting Regressor
300 trees, depth 3, lr 0.08

Output: peak glucose rise above fasting (mg/dL)

g(t) = base + rise x (t/tp) x e^(1-t/tp);
tp shifts with fat, protein, walking

Inputs (10)

Carbs, protein, fat, fiber, sleep, stress, age, BMI, fasting glucose, walk

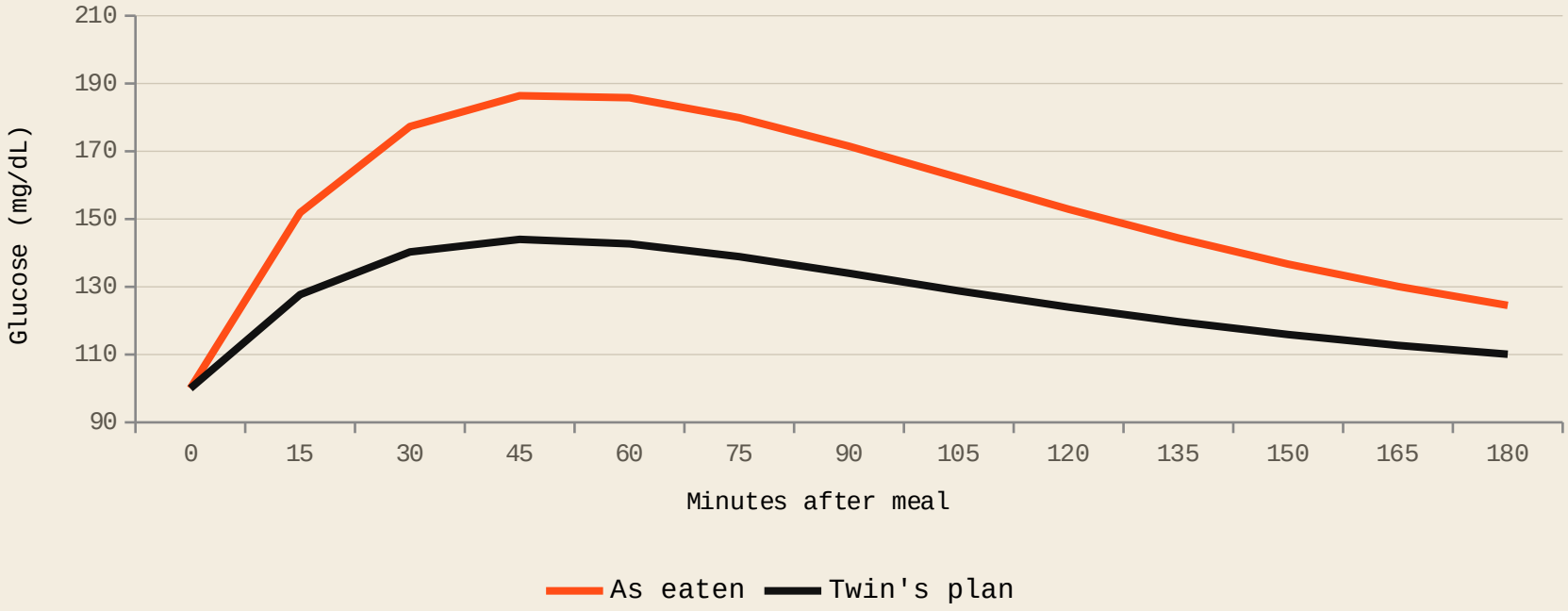
MAE

4.8 mg/dL (synthetic test set)

R2

0.97 on 20% hold-out

As eaten vs the twin's plan



Outcomes and impact

Users

Learn personal glucose triggers without needles or costly hardware.

Coaches

Test diet plans per client and explain them with curves.

Health systems

Earlier, cheaper prevention that scales through a simple API.

Privacy and limitations

- Trained on synthetic data: a proof of concept, not a clinical tool
- Not medical advice; people with diabetes should follow their doctor
- The prototype collects and stores no personal health data
- Next: validate with consented, anonymised real CGM data

What comes next

1. Calibrate with real CGM data; learn per user
2. Wearable sync: Fitbit, Apple Health, Google Fit
3. Indian food database: idli, dosa, roti, dal, rice
4. Multilingual AI coach: English, Hindi, Kannada
5. Federated learning to keep data on the device

Thank you.

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